

RREMOTE CONTROL

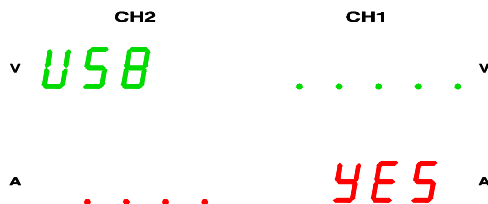
Remote Control Setup

Background	The GPD-X303S is capable of being remotely controlled via a USB connection.	
Interface		USB slave port, rear panel
COM setting	Set up the COM port inside the PC according to the following list. • Baud rate: 9600/57600 /115200 • Parity bit: None • Data bit: 8 • Stop bit: 1 • Data flow control: None	
Functionality check	Run this query command via the terminal application such as MTTTY (Multi-threaded TTY). *IDN? This should return the identification information: Manufacturer, model name, serial number, firmware version. GW INSTEK, GPD-x303S, SN: xxxxxxxx, Vx.xx	

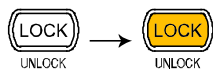
Remote Connection Step

Entering the remote control mode

1. Connect the USB cable to the slave port.
2. The connection will be automatically established, and the front panel shows a "USB...YES" message.

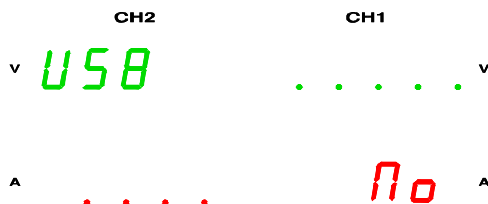


3. The power supply front panel is automatically locked (the Lock key will become activated).

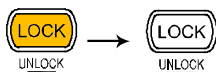


Leaving the remote control mode

1. To exit remote mode either, 1) use the LOCAL command from the terminal connection, or 2) Press the LOCK key on the front panel to return to local mode, or 3) disconnect the USB cable from the rear panel.
2. The display shows "USB...NO" message.



3. The LOCK will no longer be lit when remote mode is off.



4. The power supply goes back to the local operation mode.

Command Syntax

Command format	ISET<X>:<NR2>NL					1: command header
						2: output channel
						3: separator
						4: parameter
						5: terminator(line feed)
Output channel	1 (CH1) or 2 (CH2)					
Parameter	Type	Description	Example			
	<Boolean>	boolean logic	0 (off), 1 (on)			
	<NR1>	integers	0, 1, 2, 3			
	<NR2>	decimal numbers	0.1, 3.14, 8.5			
Terminator	Each command must end with a terminal character (new line code, ASCII: 0x0A) and each query must end with a carriage return, ASCII: 0x0D.					
Note	Commands are not case-sensitive.					

Error Messages

The following error messages might appear when the GPD-X303S cannot accept the command.

Message contents	Descriptions
a Program mnemonic too long	The command length must be 15 characters or less.
b Invalid character	Invalid characters, such as symbols, are entered. Example: VOUT#
c Missing parameter	The parameter is missing from the command. Example: VSET: (should have a number)
d Data out of range	The entered value exceeds the specification. Example: VSET:33 (should be ≤ 32V)

e	Command not allowed	The entered command is not allowed in the circumstance. Example: trying to set CH2 output while in the tracking mode.
f	Undefined header	The entered command does not exist, or the syntax is wrong.

Command List

- Detailed descriptions of each command start from the next page.
- The “HELP” command shows all the below commands and their meanings, except for the HELP command itself.

ISet<X>:<NR2>	Sets the output current.
ISet<X>?	Returns the output current setting.
VSet<X>:<NR2>	Sets the output voltage.
VSet<X>?	Returns the output voltage setting.
IOUt<X>?	Returns the actual output current.
VOUt<X>?	Returns the actual output voltage.
TRAcK<NR1>	Selects the operation mode.
BEeP<BOOLEAN>	Turn on or off the beep.
OUt<BOOLEAN>	Turn on or off the output.
STATUs?	Returns the GPD-X303S status.
*IDN?	Returns the GPD-X303S identification.
RCL<NR1>	Recalls a panel setting.
SAV<NR1>	Saves the panel setting.
HELP?	Shows the command list.
ERR?	Returns the instrument error messages.
BAUD<NR1>	Sets the baud rate.
LOCAL	Returns the instrument to local mode.

Command Details

ISET<X>:<NR2>

Description	Sets the output current for the selected channel.	
X	1= CH1, 2= CH2, (4303S: 3 = CH3, 4= CH4)	
<NR2>	Decimal number, range 0~3.200A	
Response time	Minimum 10ms	
Example	ISET1:2.234	Sets the CH1 output current to 2.234A.

ISET<X>?

Description	Returns the output current setting.	
X	1= CH1, 2= CH2, (4303S: 3 = CH3, 4= CH4)	
Response time	Minimum 10ms	
Example	ISET1?	Returns the CH1 output current setting.

VSET<X>:<NR2>

Description	Sets the output voltage.	
X	1: CH1, 2: CH2, (4303S: 3: CH3, 4: CH4)	
<NR2>	Decimal number, range 0~32.000V	
Response time	Minimum 10ms	
Example	VSET1:20.345	Sets the CH1 voltage to 20.345V

VSET<X>?

Description	Returns the output voltage setting.	
X	1: CH1, 2: CH2, (4303S: 3: CH3, 4: CH4)	
Response time	Minimum 10ms	
Example	VSET1?	Returns the CH1 voltage setting

IOUT<X>?

Description	Returns the actual output current.	
X	1: CH1, 2: CH2, (4303S: 3: CH3, 4: CH4)	
Response time	Minimum 10ms	
Example	IOUT1?	Returns the CH1 output current

VOUT<X>?

Description	Returns the actual output voltage.	
X	1: CH1, 2: CH2, (4303S: 3: CH3, 4: CH4)	
Response time	Minimum 10ms	
Example	VOUT1?	Returns the CH1 output voltage

TRACK<NR1>

Description	Selects the operation mode: independent, tracking series, or tracking parallel.	
NR1	0: Independent, 1: Series, 2: Parallel	
Response time	Minimum 10ms	
Example	TRACK0	Selects the independent mode

BEEP<Boolean>

Description	Turns the beep on or off.	
<Boolean>	0:off, 1:on	
Response time	Minimum 10ms	
Example	BEEP1	Turns on the beep

OUT<Boolean>

Description	Turns on or off the output.	
<Boolean>	0: off, 1: on	
Response time	Minimum 10ms	
Example	OUT1	Turns on the output

STATUS?

Description	Returns the GPD-X3303S.	
Response time	Minimum 10ms	

Return parameter 8 bits in the following format

Bit	Item	Description
0	CH1	0=CC mode, 1=CV mode
1	CH2	0=CC mode, 1=CV mode
2, 3	Tracking	01=Independent, 11=Tracking series, 10=Tracking parallel
4	Beep	0=Off, 1=On
5	Output	0=Off, 1=On
6, 7	Baud	00=115200bps, 01=57600bps, 10=9600bps

***IDN?**

Description	Returns the instrument identification.	
Response time	Minimum 10ms	

Return parameter GW INSTEK,GPD-X3303,SN: xxxxxxxx, Vx.xx
(Manufacturer, model name, serial number,
firmware version)

RCL<NR1>

Description	Recalls a panel setting.	
<NR1>	1 – 4: Memory 1 to 4	
Response time	Minimum 10ms	
Example	RCL1	Recalls the panel setting stored in memory 1

SAV<NR1>

Description	Stores the panel setting.	
<NR1>	1 – 4: Memory 1 to 4	
Response time	Minimum 10ms	
Example	SAV1	Stores the panel setting in memory 1

BAUD<NR1>

Description	Sets the baud rate to 9600bps/57600bps /115200bps.	
<NR1>	0: 115200bps, 1: 57600bps, 2: 9600bps	
Response time	Minimum 10ms	
Example	BAUD0	Sets the baud rate to 115200bps.

LOCAL

Description	Exits remote mode and sets the instrument to local mode.	
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Response time	Minimum 10ms
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ERR?

Description	Checks the error status of the instrument and returns the last error message.
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Response time	Minimum 10ms
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Contents	See page 46 for the list of error messages.
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HELP?

Description	Shows the command list.
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Response time	Minimum 50ms
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Return parameters	ISET<x>:<NR2> Sets the value of current. VSET<x>:<NR2> Sets the value of voltage. ISET<x> ? Return the value of current. VSET<x> ? Return the value of voltage. IOUT<x> ? Returns actual output current , VOUT<x> ? Returns actual output voltage. TRACK<NR1> Sets the output of the power supply working on independent or tracking mode. BAUD< NR1 >Set the value of baud rate. RCL<NR1> Recall the setting data from the memory which previous saved. SAV<NR1> Saves the setting data to memory. BEEP<Boolean> Sets the BEEP state on or off. OUT<Boolean> Sets the output state on or off. LOCAL Return to local mode *IDN ? Returns instrument identification. ERR ? Returns instrument error messages. STATUS ? Returns the power supply state.
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Note

All response time estimates are based on a baud rate of 115200bps. Expect longer response times with a baud rate of 57600bps or 9600bps.